In [ ]:
```python
from flask import Flask, request, jsonify
import tensorflow as tf
import numpy as np
from PIL import Image
from io import BytesIO

app = Flask(__name__)
API_KEY = "12345"

model = tf.keras.applications.MobileNetV2(weights="imagenet")

@app.route("/predict", methods=["POST"])
def predict():
    if request.headers.get("API-Key") != API_KEY:
        return jsonify({"error": "Unauthorized"}), 401

    image = Image.open(BytesIO(request.files["image"].read()))
    image = image.convert("RGB").resize((224, 224))
    image = np.array(image, dtype=np.float32)
    image = tf.keras.applications.mobilenet_v2.preprocess_input(
        np.expand_dims(image, axis=0)
    )

    prediction = model.predict(image, verbose=0)
    result = tf.keras.applications.mobilenet_v2.decode_predictions(
        prediction, top=1
    )[0][0]

    return jsonify({"prediction": result[1]})

if __name__ == "__main__":
    app.run(debug=False)
```

```
 * Serving Flask app '__main__'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deplo
yment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
127.0.0.1 - - [26/Oct/2026 09:23:50] "POST /predict HTTP/1.1" 401 -
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